

SMARTBRIDGE — Project Proposal (Proposed Solution)

Project Initialization and Planning Phase for MedSafe AI

Date:	August 12, 2026	Team ID:	MEDSAFE_AI_01
Project Title:	MedSafe AI — Healthcare Safety & Preventive Awareness Platform	Max Marks:	5 Marks

Executive Summary: The MedSafe AI proposal aims to transform personal medication safety and healthcare awareness using local AI, computer vision OCR, and fuzzy string matching. It directly tackles prescription misinterpretation, undetected adverse drug-drug interactions, and patient panic during health symptoms, ensuring strict offline privacy (100% HIPAA/GDPR compliant).

1. Project Overview

Objective	To build an air-gapped, offline-first healthcare platform that parses prescription images, checks drug interactions deterministically, and provides instant symptom triage guidance.
Scope	Covers end-to-end patient workflow: OCR handwritten/printed prescription digitization, drug lookup via RapidFuzz, local LLM safety summarization (LLaMA 3 via Ollama), and red-flag symptom risk scoring (1-10).

2. Problem Statement

Description	Patients and caregivers struggle to decode handwritten doctor prescriptions and often miss severe drug-drug interactions. Furthermore, searching symptoms online leads to panic or cloud privacy risks regarding sensitive personal health information (PHI).
Impact	Solves critical medication safety risks, prevents adverse drug events, reduces unnecessary Emergency Room visits, and guarantees complete data privacy through 100% local processing.

3. Proposed Solution

Approach	Deploy a local Python/Streamlit and FastAPI stack leveraging PyTesseract OCR for image parsing, RapidFuzz for typo-tolerant drug matching, local SQLite/JSON databases for deterministic interaction checks, and Ollama (LLaMA 3) for plain-language medical education.
Key Features	Prescription OCR Reader: Digitizes images to structured drug lists locally. Deterministic Interaction Checker: Instant severity alerts for co-prescribed meds. Symptom Triage Engine: 1-10 risk scoring with instant emergency banners for red-flag symptoms. Zero-Cloud Privacy: Complete offline operation protecting patient health data.

4. Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing, Memory & Storage Resources	8-Core CPU / NVIDIA T4 GPU (Optional), 16 GB RAM, 512 GB SSD
Software	Frameworks, Libraries & Tools	Python 3.11, Streamlit, FastAPI, PyTesseract, RapidFuzz, SQLite, Ollama (LLaMA 3 8B)
Environment	IDE & Containerization	VS Code, Docker, Docker Compose, GitHub Actions
Data	Datasets & Knowledge Bases	Local SQLite Drug Interaction Database, Local Red-Flag Emergency Keyword Dictionary